Local Evidence Is Not Completion: Evidence-Condition Geometry for Bounded Completion-Audit Workflows

Anonymous Submission

Abstract

Completion-audit workflows often stop before the remaining workload is known. Their stop signals–no-new findings, agent agreement, stable summaries, or self-reported completion–can be locally meaningful while still being unsafe as scope-wide completion certificates. The missing question is: under what evidence condition was the stop signal produced, and does that condition match the claimed scope? We study this certificate mismatch in bounded source-route audit workflows. A source-route stratum records both where the workflow searched and which audit route it applied. We use the resulting exposure geometry to build an evidence-condition controller with four modules: source-route exposure estimation, an eligibility gate, residual-evidence repair over weak plausible gaps, and a final SAFE/CONTINUE/ABSTAIN decision rule. Across four bounded completion-audit task families, homogeneous route reuse creates false certification risk; source-route granularity detects this risk where source-only control does not; broad eligibility alone is insufficient on the `urllib3` boundary; and the full controller avoids unsafe stops without becoming a never-stop rule. The result is a bounded diagnostic and control principle: completion certificates should be judged by the evidence condition that produced them, not by stop signals alone.

1 Introduction

Completion-audit workflows must eventually stop, even when the remaining workload is unknown. A local stop signal–no new findings, agreement, or a stable summary–can be true under the evidence condition that produced it while still failing as a certificate for the declared task scope. We call this *certificate mismatch*. In a code audit, for example, exhausting timeout evidence in a file does not certify TLS, retry, exception, or cleanup evidence in that same file.

We model this mismatch with *evidence-condition geometry*. A source-route stratum records both where a workflow searched and which audit route it applied. The exposure distribution over these strata is a runtime certificate diagnostic: it tells us whether a proposed stop is supported by broad source-route evidence or by a localized condition. The diagnostic does not know the hidden workload and does not prove open-world completion.

We build a controller around this diagnostic. It estimates source-route exposure, checks eligibility, repairs weak plausible strata, and returns SAFE/CONTINUE/ABSTAIN. A generic verifier gate can avoid unsafe SAFE decisions by failing closed, but it gives no source-route diagnosis or repair target. Our controller aims for a stronger operational property: safe decisions when the condition is defensible, actionable CONTINUE decisions when residual evidence is found, and explicit ABSTAIN when no certificate can be formed. The contribution is therefore not a new retrieval policy alone; it is a certificate interface for deciding whether a stop proposal is safe, actionable, or still unsupported.

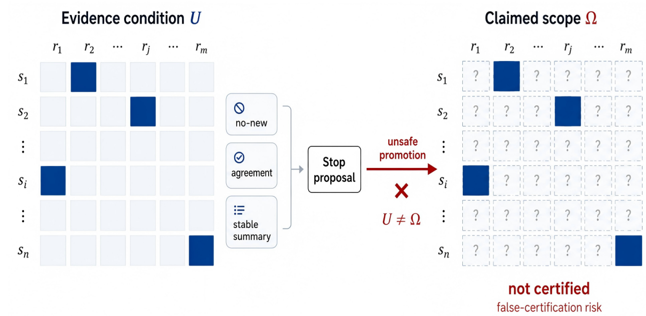

Figure 1: Certificate mismatch from local stop evidence. A workflow may observe no-new findings, agreement, or a stable summary under evidence condition U. Promoting that stop proposal to a completion claim over Ω is unsafe when $U \neq \Omega$; the unobserved source-route strata are not certified.

Experiments on two generated audits and two frozen real-repository audits test this safety–actionability–cost tradeoff. Homogeneous route reuse creates false certification risk; source-route geometry exposes risks hidden by source-only coverage; the `urllib3` boundary shows that eligibility is not proof; and the full controller avoids unsafe SAFE decisions while turning residual-positive unsafe states into actionable CONTINUE rather than merely abstaining. The paper makes three claims. First, stop evidence has a condition, and local conditions cannot certify broader source-route scopes. Second, source-route geometry is a useful stop-time warning signal because it separates file coverage from route coverage. Third, a repair-aware controller is more useful than a generic fail-closed verifier because it preserves safety while producing the next audit target when residual evidence appears.

The claim is bounded: completion certificates should be judged by the evidence condition that produced them, not by stop signals alone.

2 Related Work

Our work sits between three literatures that usually treat stopping, evidence, and verification separately. High-recall review, total recall, active search, and review stopping rules ask how to retrieve more positives, estimate remaining mass, or decide whether a review has saturated under budget [Grossman and Cormack, 2011, Cormack and Grossman, 2014, 2015, 2016, Grossman et al., 2016, Roegiest et al., 2015, Voorhees, 2002, Wallace et al., 2010, Lewis and Gale, 1994, Settles, 2009, Garnett et al., 2012, Attenberg et al., 2015, Good, 1953, Chao, 1984, Chao et al., 2014]. Those methods are closest in spirit to our concern with false stopping, but they often reason with review protocols, sampling assumptions, prevalence estimates, or post-hoc recall targets. Our controller is deliberately poorer at runtime: it is not given oracle totals, hidden missing mass, post-hoc recall, or unseen true-item counts. It therefore asks a narrower certificate question: whether the observed evidence condition is broad enough to support the declared source-route scope of the completion claim.

A second line of work studies tool-using, web, code, and multi-agent systems that allocate work across searches, tools, memories, repositories, and intermediate states [Yao et al., 2023, Schick et al., 2023, Nakano et al., 2021, Qin et al., 2024, Wu et al., 2023, Park et al., 2023, Wang et al., 2024, Deng et al., 2024, Zhou et al., 2024, Liu et al., 2024, Yao et al., 2022, Jimenez et al., 2024, Shinn et al., 2024, Madaan et al., 2023]. These systems make our failure mode more common: a workflow may cover many sources but only through a narrow route, or may repeat one local route until the transcript appears stable. Benchmarks and factuality systems then evaluate final task success, answer support, or post-hoc correctness [Hendrycks et al., 2021a, Liang et al., 2022, Srivastava et al., 2023, Lin et al., 2022, Thorne et al., 2018, Gao et al., 2023, Min et al., 2023, Schuster et al., 2021, Gehrmann et al., 2021]. We instead evaluate the stop certificate itself: what condition produced it, and what action should follow when that condition is weak.

Finally, verification and oversight methods can reject unsupported outputs or fail closed when a checker reports unresolved warnings [Dietterich, 2000, Cobbe et al., 2021, Wang et al., 2023, Amodei et al., 2016, Christiano et al., 2017, Irving et al., 2018, Leike et al., 2018, Hendrycks et al., 2021b]. We include a generic verifier-gate for exactly this reason. Its limitation is not that abstention is unsafe; abstention is often the right conservative answer. Its limitation is that a warning-only gate does not say which source-route condition is missing or where the next audit should go. Evidence-condition control treats the stop claim as a structured certificate: diagnose the source-route geometry, repair weak plausible strata, and return SAFE, CONTINUE, or ABSTAIN with an operational meaning.

3 Problem Setting

We study bounded completion-audit workflows with unknown residual workload and declared source-route scope. A workflow receives a task, allocates work across agents or tools, gathers evidence under a finite budget, and must decide whether the task is complete within that declared scope. Let

$$W = (T, S, R, A, B),$$

where T is the task, S is a set of sources, R is a set of routes, A is the set of agents or tools, and B is a finite budget. A source is a bounded evidence region, such as a file, module, document, source family, or repository component. A route is an audit or search lens applied to a source.

At runtime, the workflow produces evidence events

$$e_t = (a_t, x_t, r_t, u_t, o_t, c_t),$$

where a_t is the agent or tool, $x_t \in S$ is the source, $r_t \in R$ is the route, u_t is the action, o_t is the observation, and c_t is the cost. Every evidence event is produced under a source-route condition. A source-route stratum is $s = (x, r)$, and the intended evidence-condition space is

$$\Omega = S \times R.$$

The source-route space does not require the workflow to know all true missing items in advance. It requires the intended audit scope to be defined: which sources and routes are relevant to the completion claim. This assumption is weaker than knowing the answer set and stronger than an unbounded open-world search. It matches many practical audits: a system may know which repositories, documents, modules, or evidence sources are in scope, and it may know which audit routes are relevant, while still not knowing how many true findings remain. The controller therefore evaluates the condition under which evidence was gathered, not the unknown total workload itself.

A stop claim is a workflow assertion

$$C_t : \quad T \text{ is complete over the intended workload scope.}$$

A false certification occurs when the workflow accepts SAFE for a stop claim that is not supported by the evaluation oracle:

$$\text{SAFE}(C_t) = \text{true}, \qquad \text{complete}(T) = \text{false}.$$

In item-discovery subclasses, this is operationalized as accepting SAFE while bounded-oracle recall is below the completion threshold. In broader completion audits, it means accepting a global completion claim while relevant support, contradiction, or unresolved evidence remains outside the evidence condition. The oracle is used only for evaluation. It is not available to route assignment, repair scoring, eligibility checks, or stop decisions.

4 Evidence-Condition Geometry

Let $v_t(s)$ be the runtime-visible exposure count for source-route stratum s up to time t. Exposure may count visits, scans, route-specific audits, tool calls, or other logged search actions. Define

$$p_{\exp}(t, s) = \frac{v_t(s)}{\sum_{s' \in \Omega} v_t(s')}.$$

This distribution is a point on the source-route coverage simplex. It describes where and how completion evidence was produced.

We use two runtime summaries:

$$\mathrm{support}(t) = \frac{|\{s : v_t(s) > 0\}|}{|\Omega|},$$

$$G_{\exp}(t) = \mathrm{Gini}(\{v_t(s) : s \in \Omega\}).$$

These summaries are operational diagnostics, not universal laws. They summarize whether the evidence condition is localized or broad enough to make a completion certificate eligible for consideration.

Proposition 1 (Local no-new evidence is not a global certificate)**.** *Let $U \subset \Omega$ be a strict subset of the intended source-route space. Suppose all evidence used to support a stop claim C_t is produced inside U, and suppose the workflow observes no-new evidence only for actions inside U. Then the no-new evidence can support local exhaustion over U, but it cannot by itself support global completion over Ω.*

Proof sketch. No-new evidence is conditioned on the source-route actions that were executed. If no events are produced in $\Omega \setminus U$, the workflow has not observed the absence of residual evidence in those unexposed strata. A world in which U is exhausted and $\Omega \setminus U$ contains residual evidence is observationally indistinguishable from a globally complete world under the workflow's local evidence log. Therefore local no-new evidence rules out additional findings only under the searched condition U; without additional assumptions about unsearched strata, it cannot rule out residual evidence in $\Omega \setminus U$.

Source-only geometry is too coarse: auditing a file for timeout behavior does not certify that TLS, retry, exception, or cleanup behavior has been audited in that file. Source-route-action geometry is more detailed, but it is sensitive to logging conventions and current experiments do not show a stable advantage over source-route. Source-route is the default because it captures both where and how evidence was produced while remaining interpretable and runtime-computable. The geometry is thus a certificate diagnostic rather than a semantic theory of all possible evidence. It asks whether the current log covers the intended source-route space broadly enough for the proposed stop claim; it does not assert that all routes are equally important or that support/Gini thresholds transfer unchanged across domains.

5 Method

The controller has four modules, shown in Algorithm 1. *Exposure estimation* maps the runtime evidence log to source-route counts $v_t(s)$, support $\mathrm{support}(t)$, and Gini concentration $G_{\exp}(t)$. *Eligibility* checks whether the stop condition is broad enough to consider a certificate. *Residual repair* probes weak plausible strata using only runtime-visible signals. The final rule returns SAFE only when the evidence condition is eligible and repair is nonproductive; residual evidence yields actionable CONTINUE, and unresolved ineligibility yields ABSTAIN.

Algorithm 1 Evidence-Condition Controller

Require: log E_t, space Ω, claim C_t, budget B, thresholds τ_s, τ_g, repair rule $\mathcal{R}$
Ensure: decision in $\{\textsc{Safe}, \textsc{Continue}, \textsc{Abstain}\}$
Compute $v_t(s)$, $p_{\exp}(t,s)$, $\mathrm{support}(t)$, and $G_{\exp}(t)$
$\mathit{eligible} \leftarrow [\mathrm{support}(t) \geq \tau_s \wedge G_{\exp}(t) \leq \tau_g]$
if not $\mathit{eligible}$ and $B = 0$ **then**
  return ABSTAIN
end if
if not $\mathit{eligible}$ or weak plausible gaps remain **then**
  Audit $Q = \mathcal{R}(E_t, \Omega, C_t)$ within B
  if residual evidence appears **then**
    return CONTINUE
  end if
  Recompute $\mathit{eligible}$
end if
if $\mathit{eligible}$ and no residual evidence appeared **then**
  return SAFE
end if
return ABSTAIN

The repair instance used in experiments ranks strata by

$$\mathrm{priority}(s) = \mathrm{under\_exposure}(s) \cdot \mathrm{runtime\_potential}(s).$$

The potential term may use source text, route names, route match counts, source size, or ledger-visible notes. It may not use oracle labels, oracle totals, post-hoc recall, hidden missing mass, or undiscovered true-item counts. Residual-potential is a mechanism-aligned probe, not an optimal active-search claim. Thus the method's value is stop-time certificate diagnosis plus targeted repair, not universal completion proof.

Proposition 2 (Controller certificate semantics)**.** *For a declared source-route scope Ω, thresholds τ_s, τ_g, and a fixed repair rule $\mathcal{R}$ that uses only runtime-visible signals, Algorithm 1 returns* SAFE *only if (i) the exposure condition satisfies the eligibility gate and (ii) repair or audit reveals no residual evidence within the allocated repair budget. If residual evidence is found, the returned decision is* CONTINUE*; if eligibility cannot be established under budget, the returned decision is* ABSTAIN.

The proof sketch is in the supplement; the proposition is about the controller's declared semantics, not about absence of all unobserved evidence.

6 Experiments and Results

We evaluate whether a stop controller is *safe, actionable, and cost-aware* under a declared source-route scope. Each comparison fixes the workflow trajectory before scoring: a policy sees only runtime-visible evidence, optional repair scans a configured budget, and oracle labels are revealed only after the decision. Thus the experiments test completion con-

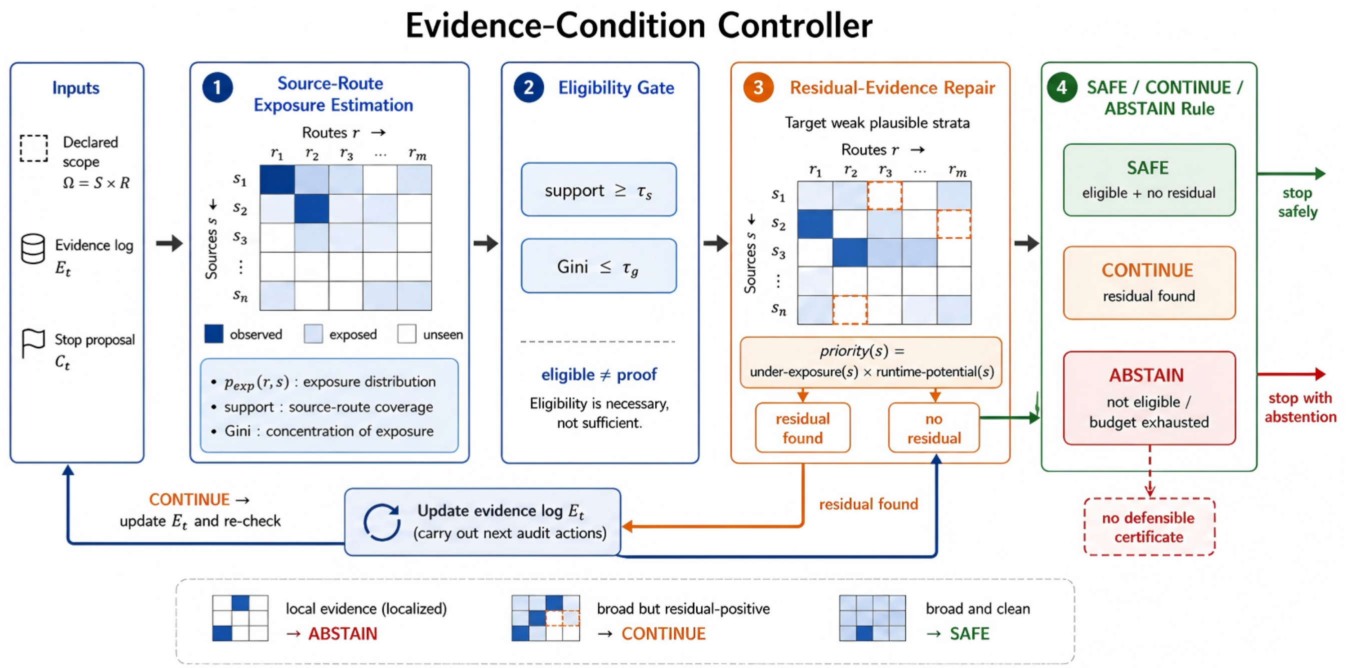

Figure 2: Evidence-condition controller. The controller maps a runtime evidence log, declared source-route scope, and stop proposal to an exposure distribution over the intended source-route space. The support/Gini check is an eligibility test, not a completion proof. If the condition is localized or weak plausible gaps remain, the controller repairs or continues. It returns SAFE only when the evidence condition is broad and repair or audit reveals no residual evidence; otherwise it returns CONTINUE or ABSTAIN.

trol rather than oracle-assisted discovery. We use two generated hidden-oracle audits (`policy-docset`, 24 clauses over four routes; `code-repo`, 20 risks over three routes) and two frozen repository audits (`requests`, 298 pattern-defined items; `urllib3`, 699 pattern-defined items). The repository oracles are useful scoring mechanisms, not human semantic gold labels.

The global protocol is deliberately fixed. Runtime rules may use evidence logs, source text, route names, source sizes, and ledger-visible notes; they may not use oracle labels, oracle totals, post-hoc recall, or undiscovered true-item counts. We report FCR (unsafe states certified SAFE), safe coverage (complete states certified SAFE), actionability (CONTINUE on unsafe states with residual candidates), and post-stop repair cost. Config files specify seeds, routes, thresholds, budgets, oracle paths, and output paths; full command lines, CSV/JSON exports, and denominator tables are in the supplement.

Local stop evidence causes false certification. We first ask whether ordinary local stop signals are enough for a scope-wide completion claim. Homogeneous-route stops touch sources through one repeated route; if accepted as SAFE, they falsely certify all four task families, with bounded-oracle recall as low as 0.104 on `requests`. The result is the basic failure mode: a no-new-local-finding stop can be true under its local evidence condition while still being the wrong certificate for global completion.

Source-only coverage hides route mismatch. The granularity ablation asks whether file/source coverage is enough. At the same homogeneous stops, source-only support is 1.0 for every task, while source-route support is only 0.20–0.33 and recall is 0.104–0.708. Source-only coverage therefore creates an illusion of completion: touching a file through a timeout route does not certify TLS, retry, exception, cleanup, or policy routes in that file.

Source-route geometry exposes the mismatch. Figure 3 next plots source-route support against exposure Gini. Homogeneous stops remain low-support and high-concentration; route- partitioned and extended audits broaden support and reduce concentration. The claim is diagnostic, not absolute: source-route geometry exposes a weak stop condition that scalar source coverage misses.

Eligibility is only a precondition. `urllib3` supplies the boundary case. Its route-partitioned state covers 24/30 strata and passes the eligibility gate, but runtime-plausible residual strata remain; the full controller returns CONTINUE, not SAFE. Post-hoc scoring explains the refusal: support is 0.800, Gini 0.647, recall 0.835, and 115 oracle items remain missed, while the extended audit reaches support and recall 1.000. This denominator is one fixed eligible-but-residual-positive `urllib3` state, not the seeded controller validation denominator used below.

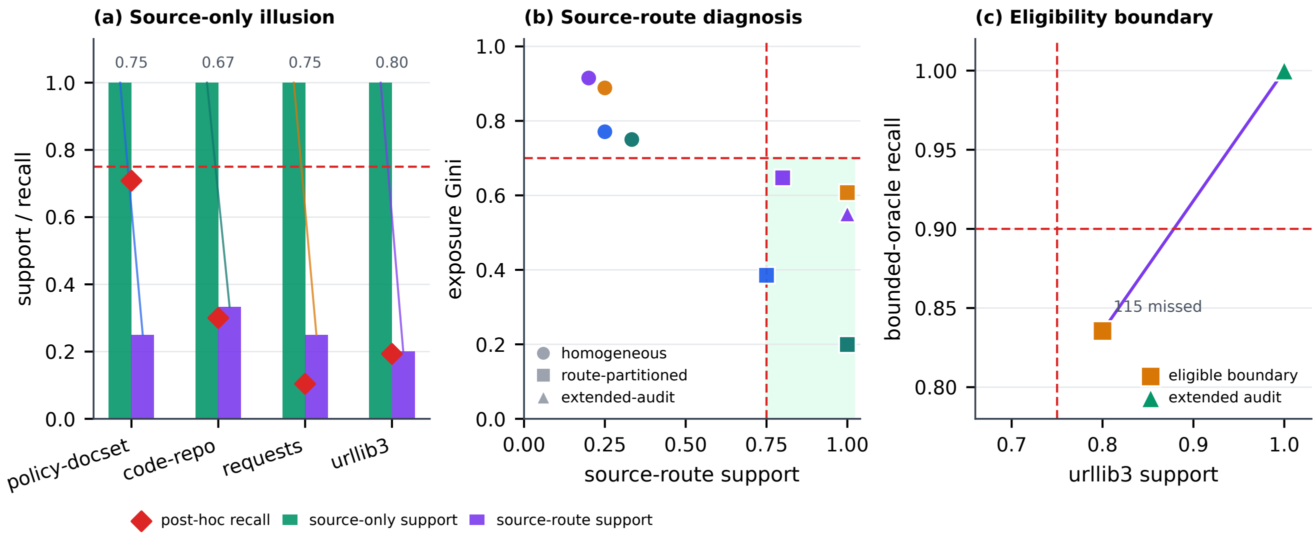

Figure 3: Diagnostic evidence for the stop condition. Source-only exposure looks complete at localized stops, but source-route support reveals route mismatch and low post-hoc recall; numbers above panel (a) are the source-only minus source-route support gaps. The support–Gini view shows how route-partitioned and extended audits move toward the configured eligibility region. The `urllib3` boundary is one fixed eligible-but-residual-positive state: support 0.800, recall 0.835, and 115 missed oracle items before the extended audit reaches full support and recall.

Full control is safe and actionable. We then compare Naive, Source-only, Verifier-gate, Eligibility-only, and Full on the same seeded unsafe repair states and seeded complete-state perturbations. Naive and Source-only certify all 400 unsafe states; Verifier-gate and Eligibility-only avoid unsafe SAFE by returning 400 ABSTAIN decisions; the Full controller also has FCR 0 and preserves $1200/1200$ complete-state SAFE decisions, but returns 400 CONTINUE decisions on productive unsafe states. Thus the advantage over a generic verifier is actionability: ABSTAIN refuses certification, whereas CONTINUE points to the next audit target.

Residual repair gives gain at audit cost. Finally, we vary repair targets. Residual-potential finds 253.0 new oracle items at mean cost 4275.5; random repair finds 69.0 at cost 3481.1; and high-potential finds 226.0 at cost 3824.0. Repair is therefore the cost-bearing mechanism that turns a weak stop into a next action; it is a high-yield heuristic here, not an optimal active-search claim. Threshold/budget sweeps and a small claim-verification interface sanity check are reported in the supplement.

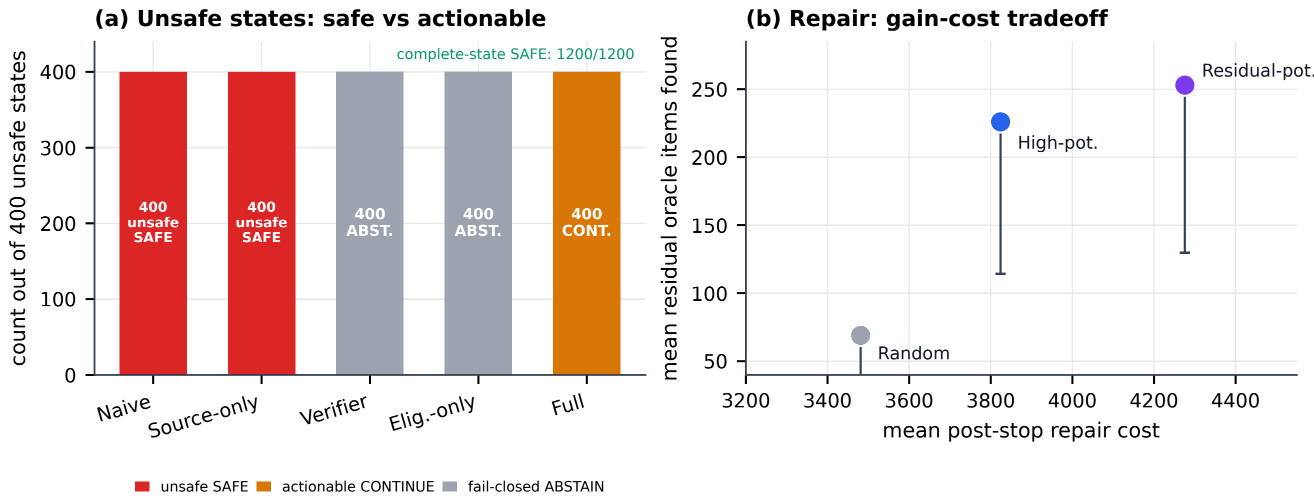

Figure 4: Controller evaluation on seeded validation states. Left: on 400 seeded unsafe residual-positive states, Verifier-gate is safe but fail-closed (ABSTAIN), while the Full controller is safe and actionable (CONTINUE); all policies certify $1200/1200$ seeded complete states as SAFE. Right: repair targeting buys residual evidence at explicit audit cost. These seeded denominators differ from the one-state `urllib3` boundary in Figure 3.

7 Discussion and Limitations

The positive result is bounded but operationally useful. A SAFE decision certifies the declared source-route condition under the configured budget and runtime-visible signals; it is not a universal guarantee that no further facts exist. In practice, support and Gini are stop-time warning signals, threshold and budget sweeps set conservatism, and residual repair turns a weak certificate into a next audit target. The limitations are real: external repository oracles are pattern-defined over frozen snapshots, residual-potential repair is a heuristic rather than optimal active search, and the claim-verification pilot is only an interface sanity check.

8 Conclusion

Local stop signals are not completion certificates. Source-route exposure geometry makes the evidence condition of a stop proposal visible; residual repair then turns uncertainty into SAFE, actionable CONTINUE, or conservative ABSTAIN. The contribution is intentionally bounded, but it closes a practical loop: judge completion claims by the condition that produced them, not by stop signals alone.

References

Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in ai safety. *arXiv preprint arXiv:1606.06565*, 2016.

Josh Attenberg, Panos Ipeirotis, and Foster Provost. Beat the machine: Challenging humans to find a predictive model's unknown unknowns. *Journal of Data and Information Quality*, 6(1), 2015.

Anne Chao. Nonparametric estimation of the number of classes in a population. *Scandinavian Journal of Statistics*, 11(4):265–270, 1984.

Anne Chao, Nicholas J. Gotelli, T. C. Hsieh, Elizabeth L. Sander, K. H. Ma, Robert K. Colwell, and Aaron M. Ellison. Rarefaction and extrapolation with hill numbers: A framework for sampling and estimation in species diversity studies. *Ecological Monographs*, 84(1):45–67, 2014.

Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, 2017.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Gordon V. Cormack and Maura R. Grossman. Evaluation of machine-learning protocols for technology-assisted review in electronic discovery. In *Proceedings of the 37th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 153–162, 2014.

Gordon V. Cormack and Maura R. Grossman. Multi-faceted recall of continuous active learning for technology-assisted review. In *Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2015.

Gordon V. Cormack and Maura R. Grossman. Engineering quality and reliability in technology-assisted review. *Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 75–84, 2016.

Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Sam Stevens, Boshi Wang, Huan Sun, and Yu Su. Mind2web: Towards a generalist agent for the web. In *Advances in Neural Information Processing Systems*, 2024.

Thomas G. Dietterich. Ensemble methods in machine learning. *Multiple Classifier Systems*, pages 1–15, 2000.

Luyu Gao, Zhuyun Dai, Panupong Pasupat, Anthony Chen, Arun Tejasvi Chaganty, Yicheng Fan, Vincent Y. Zhao, Ni Lao, Hongrae Lee, Da-Cheng Juan, and Kelvin Guu. Rarr: Researching and revising what language models say, using language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023.

Roman Garnett, Yamuna Krishnamurthy, Xuehan Xiong, Jeff Schneider, and Richard Mann. Bayesian optimal active search and surveying. In *Proceedings of the 29th International Conference on Machine Learning*, pages 843–850, 2012.

Sebastian Gehrmann, Tosin Adewumi, Karmanya Aggarwal, Pranav Ammanamanchi, Aremu Anuoluwapo, Antoine Bosselut, Khyathi Raghavi Chandu, Miruna Clinciu, Dipanjan Das, Kaustubh Dhole, Wanyu Du, Esin Durmus, Ondrej Dušek, Chris Emezue, Varun Gangal, Cristina Garbacea, Tatsunori Hashimoto, Yufang Hou, Yacine Jernite, Harsh Jhamtani, Yangfeng Ji, Shailza Jolly, Mihir Kale, Jungo Kasai, Daniel Khashabi, Venelin Kovatchev, Kalpesh Krishna, Alok Kumar, Faisal Ladhak, Yi Luan, Mounica Maddela, Khyathi Mahajan, Simon Mille, Moin Nadeem, Shashi Narayan, Vitaly Nikolaev, Tong Niu, Laura Perez-Beltrachini, Denis Peskov, Shrimai Prabhumoye, Abhinav Rastogi, Anna Rogers, João Sedoc, Thibault Sellam, Diptesh Kanojia Shah, Anastasia Shimorina, Yixuan Su, Alex Tang, Yi-Lin Tian, Ronen Toledo, Chris van der Lee, Siddharth Vashishtha, Thuy-Trang Vu, Wei Xu, Diyi Yang, Jie Zhao, and Chunting Zhou. The gem benchmark: Natural language generation, its evaluation and metrics. *Transactions of the Association for Computational Linguistics*, 9: 137–160, 2021.

I. J. Good. The population frequencies of species and the estimation of population parameters. *Biometrika*, 40(3–4): 237–264, 1953.

Maura R. Grossman and Gordon V. Cormack. Technology-assisted review in e-discovery can be more effective and more efficient than exhaustive manual review. *Richmond Journal of Law and Technology*, 17(3), 2011.

Maura R. Grossman, Gordon V. Cormack, and Adam Roegiest. Trec 2015 total recall track overview. In *Proceedings of The Twenty-Fourth Text REtrieval Conference*, 2016.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021a.

Dan Hendrycks, Nicholas Carlini, John Schulman, and Jacob Steinhardt. Unsolved problems in ml safety. *arXiv preprint arXiv:2109.13916*, 2021b.

Geoffrey Irving, Paul Christiano, and Dario Amodei. Ai safety via debate. *arXiv preprint arXiv:1805.00899*, 2018.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. Swe-bench: Can language models resolve real-world github issues? In *International Conference on Learning Representations*, 2024.

Jan Leike, David Krueger, Tom Everitt, Miljan Martic, Vishal Maini, and Shane Legg. Scalable agent alignment via reward modeling: A research direction. *arXiv preprint arXiv:1811.07871*, 2018.

David D. Lewis and William A. Gale. A sequential algorithm for training text classifiers. In *Proceedings of the 17th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 3–12, 1994.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Connor Cosgrove, Christopher D. Manning, Christopher Ré, Diana Acosta-Navas, Drew A. Hudson, Eric Zelikman, Esin Durmus, Faisal Ladhak, Frieda Rong, Hongyu Ren, Huaxiu Yao, Jue Wang, Keshav Santhanam, Laurel Orr, Lucia Zheng, Mert Yuksekgonul, Mirac Suzgun, Nathan Kim, Neel Guha, Niladri Chatterji, Omar Khattab, Peter Henderson, Qian Huang, Ryan Chi, Sang Michael Xie, Shibani Santurkar, Surya Ganguli, Tatsunori Hashimoto, Thomas Icard, Tianyi Zhang, Vishrav Chaudhary, William Wang, Xuechen Li, Yifan Mai, Yuhui Zhang, and Yuta Koreeda. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.

Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, 2022.

Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, and Jie Tang. Agentbench: Evaluating llms as agents. In *International Conference on Learning Representations*, 2024.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Sean Welleck, Bodhisattwa Prasad Majumder, Shashank Gupta, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems*, 2023.

Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wentau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023.

Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, Xu Jiang, Karl Cobbe, Tyna Eloundou, Gretchen Krueger, Kevin Button, Matthew Knight, Benjamin Chess, and John Schulman. Webgpt: Browser-assisted question-answering with human feedback. *arXiv preprint arXiv:2112.09332*, 2021.

Joon Sung Park, Joseph C. O'Brien, Carrie J. Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.

Yujia Qin, Shengding Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Yulong Hong, Ruobing Tian, Ruixue Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, and Maosong Sun. Toolllm: Facilitating large language models to master 16000+ real-world apis. In *International Conference on Learning Representations*, 2024.

Adam Roegiest, Gordon V. Cormack, and Maura R. Grossman. Trec 2015 total recall track guidelines and assessment process. In *Proceedings of The Twenty-Fourth Text REtrieval Conference*, 2015.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, 2023.

Tal Schuster, Adam Fisch, and Regina Barzilay. Get your vitamin c! robust fact verification with contrastive evidence. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, 2021.

Burr Settles. Active learning literature survey. Technical Report 1648, University of Wisconsin–Madison, 2009.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, 2024.

Aarohi Srivastava et al. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. In *Transactions on Machine Learning Research*, 2023.

James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. Fever: A large-scale dataset for fact extraction and verification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics*, 2018.

Ellen M. Voorhees. Overview of the trec 2002 question answering track. In *Proceedings of the Eleventh Text REtrieval Conference*, 2002.

Byron C. Wallace, Thomas A. Trikalinos, Joseph Lau, Carla Brodley, and Christopher H. Schmid. Active learning for biomedical citation screening. In *Proceedings of the 16th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2010.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. In *Transactions on Machine Learning Research*, 2024.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.

Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed H. Awadallah, Ryen W. White, Doug Burger, and Chi Wang. Autogen: Enabling next-gen llm applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155*, 2023.

Shunyu Yao, Rohan Rao, Matthew Hausknecht, and Karthik Narasimhan. Alfworld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations*, 2022.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023.

Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. Webarena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations*, 2024.